Question 1

1a. Let

$$y = \sin^{-1} x \rightarrow x = \sin y$$ [1 mark]

Differentiate both sides with respect to x:

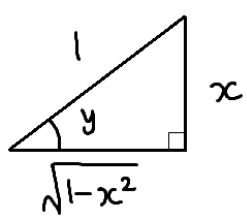
$$1 = \cos y \cdot \frac{\mathrm{d}y}{\mathrm{d}x} \rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos y} = \frac{1}{\cos \sin^{-1} x}$$ [2 marks]

Use identity: $\sin^2 x + \cos^2 x = 1 \rightarrow \cos x = \pm\sqrt{(1 - \sin^2 x)}$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \pm\frac{1}{\sqrt{1 - \sin^2 \sin^{-1} x}} = \pm\frac{1}{\sqrt{1 - x^2}}$$ [1 mark]

$y = \sin^{-1} x$ is an increasing function so gradient is always positive → reject -ve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1 - x^2}}.$$ [1 mark]



(Could have also drawn a simple right-triangle to avoid the identity: so $\cos y = 1 / \sqrt{(1 - x^2)}$ directly.
Still must justify rejecting -ve solution:
$|y| \leq \pi/2 \rightarrow \cos y > 0$, or similar.)

1b.

i) Starting with

$$f(x) = x\sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a}$$

The first term uses the product rule. To differentiate $\sqrt{(a^2 - x^2)}$, write as $(a^2 - x^2)^{1/2}$ and use chain rule, which gives $\frac{1}{2}(a^2 - x^2)^{-1/2}(-2x)$.

The second term uses the previous result in part a, with a factor of 1/a required from the chain rule. This produces

$$f'(x) = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} + \frac{a}{\sqrt{1 - \left(\frac{x}{a}\right)^2}}$$

[4 marks; 2 for first two terms, 2 for last term]

Multiply the top and bottom of the third term by a = $\sqrt{a^2}$:

$$f'(x) = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} + \frac{a^2}{\sqrt{a^2 - x^2}}$$ [1 mark]

Factor out $1/\sqrt{(a^2 - x^2)}$ from the second and third terms:

$$f'(x) = \sqrt{a^2 - x^2} + \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} = \sqrt{a^2 - x^2} + \sqrt{a^2 - x^2} = 2\sqrt{a^2 - x^2}.$$

[1 mark]

ii) Equation of circle: $x^2 + y^2 = a^2 \rightarrow y = \sqrt{(a^2 - x^2)}$ [1 mark]
(positive solution only since semi-circle is above x-axis)
Region is symmetric about y-axis so the area can be written as

$$\text{area} = 2\int_0^{a/2} \sqrt{a^2 - x^2}\ \mathrm{d}x = \int_0^{a/2} 2\sqrt{a^2 - x^2}\ \mathrm{d}x$$ [1 mark]

This is the same as the result in b.i), so

$$\text{area} = \int_0^{a/2} f'(x)\ \mathrm{d}x = f\left(\frac{a}{2}\right) - f(0)$$ [1 mark]

Putting x = 0 into f(x), we find f(0) = 0, so the area is f(a/2). [1 mark]

iii) Area of semicircle = $\frac{1}{2}\pi a^2$ [1 mark]
Area of shaded region = f(a/2) = (a/2) * $\sqrt{(3a^2/4)} + a^2 \sin^{-1} \frac{1}{2}$
= $\sqrt{3}/4\ a^2 + 1/6\ \pi a^2$ [1 mark]
Ratio = $(\sqrt{3}/4 + 1/6\ \pi) / (1/2\ \pi)$

$$= \frac{\sqrt{3}}{2\pi} + \frac{1}{3}.$$ [1 mark]

Question 2

2a. From the definition of the derivative as a limit, we start with

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot x = \lim_{h\to 0}\frac{\cot(x+h)-\cot x}{h}$$ [1 mark]

Write cot *x* = 1 / (tan *x*):

$$= \lim_{h\to 0}\frac{\frac{1}{\tan(x+h)}-\frac{1}{\tan x}}{h}$$ [1 mark]

Use addition formula: tan(x + h) = (tan x + tan h) / (1 - tan x tan h)

$$= \lim_{h\to 0}\frac{\frac{1-\tan x\tan h}{\tan x+\tan h}-\frac{1}{\tan x}}{h}$$ [1 mark]

Subtract fractions with common denominator: [2 marks]

$$= \lim_{h\to 0}\frac{\frac{\tan x(1-\tan x\tan h)}{\tan x(\tan x+\tan h)}-\frac{\tan x+\tan h}{\tan x(\tan x+\tan h)}}{h} = \lim_{h\to 0}\frac{\tan x-\tan^2 x\tan h-\tan x-\tan h}{h\tan x(\tan x+\tan h)}$$

$$= \lim_{h\to 0}\frac{-\tan^2 x\tan h-\tan h}{h\tan x(\tan x+\tan h)} = -\lim_{h\to 0}\frac{\tan h(1+\tan^2 x)}{h\tan x(\tan x+\tan h)} = -\lim_{h\to 0}\frac{\tan h\sec^2 x}{h\tan x(\tan x+\tan h)}$$

Since *h* is small, apply the small-angle approximation: tan *h* ≈ *h*:

$$= -\lim_{h\to 0}\frac{h\sec^2 x}{h\tan x(\tan x+h)} = -\lim_{h\to 0}\frac{\sec^2 x}{\tan x(\tan x+h)}$$ [1 mark]

As *h* approaches zero, this expression becomes

$$= -\frac{\sec^2 x}{\tan^2 x} = -\frac{1}{\cos^2 x}\div\frac{\sin^2 x}{\cos^2 x} = -\frac{1}{\sin^2 x} = -\csc^2 x.$$ [1 mark]

2b. Let

$$y = \cot^{-1} x \to x = \cot y$$ [1 mark]

Differentiating both sides with respect to x,

$$1 = -\csc^2 y \cdot \frac{\mathrm{d}y}{\mathrm{d}x}$$ [1 mark]

Multiply both sides by $\sin^2 y$ to isolate dy/dx:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin^2 y$$ [1 mark]

Put expression back in:

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot^{-1} x = -\sin^2 \cot^{-1} x = -\frac{1}{\csc^2 \cot^{-1} x} = -\frac{1}{1+\cot^2 \cot^{-1} x} = -\frac{1}{1+x^2}$$

[1 mark]

(Alternatively, could have put $\cot^{-1} x = \tan^{-1}(1/x)$ and used the geometric reasoning similar to part 1a)

2c. Since cot x = 1/tan x, we can write (for positive arguments) $\cot^{-1} x = \tan^{-1} 1/x$, so the function in terms of inverse cotangents is

$$y = \cot^{-1} x^2 + \cot^{-1}(x-a)^2$$

Differentiate w.r.t. x, using chain rule and previous result:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2x}{1+x^4} - \frac{2(x-a)}{1+(x-a)^4}$$

[1 mark]

Stationary point implies dy/dx = 0, so set

$$\frac{2x}{1+x^4} + \frac{2(x-a)}{1+(x-a)^4} = 0$$

[1 mark]

Combine fractions and expand:

$$2x(1+(x-a)^4) + 2(x-a)(1+x^4) = 0$$
$$x(1+x^4-4ax^3+6a^2x^2-4a^3x+a^4) + (x-a)(1+x^4) = 0$$
$$(x^5-4ax^4+6a^2x^3-4a^3x^2+(a^4+1)x) + (x+x^5-a-ax^4) = 0$$
$$2x^5-5ax^4+6a^2x^3-4a^3x^2+(a^4+2)x-a = 0.$$

[3 marks]

(Despite being a 5th-degree polynomial with an unknown constant, this equation can be solved - by factoring out (x - a/2) and exploiting the symmetry of the function with a substitution u = x - a/2, all stationary points can be found exactly. The condition for having one stationary point can be derived from this.)

2d. Find second derivative (quotient and chain rule):

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2x}{1+x^4} - \frac{2(x-a)}{1+(x-a)^4} \rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{2(1+x^4)-8x^4}{(1+x^4)^2} - \frac{2+2(x-a)^4-8(x-a)^4}{(1+(x-a)^4)^2}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{6x^4-2}{(1+x^4)^2} + \frac{6(x-a)^4-2}{(1+(x-a)^4)^2}$$

[2 marks]

Substitute in x = a/2:

$$f''\left(\frac{a}{2}\right) = \frac{\frac{3a^4}{8}-2}{(1+\frac{a^4}{16})^2} + \frac{\frac{3a^4}{8}-2}{(1+\frac{a^4}{16})^2} = \frac{\frac{3a^4}{4}-4}{(1+\frac{a^4}{16})^2} = \frac{64(3a^4-16)}{(a^4+16)^2}$$

[2 marks]

Using the given range of *a* implies

$$|a| \leq \frac{2}{\sqrt[4]{3}} \rightarrow a^2 \leq \frac{4}{\sqrt{3}} \rightarrow a^4 \leq \frac{16}{3} \rightarrow 3a^4 - 16 \leq 0$$

[1 mark]

and so the numerator of f''(a/2) is negative or zero.

When $|a| < \frac{2}{\sqrt[4]{3}}$, the second derivative is negative, so it is a **maximum**.

[1 mark]

When $|a| = \frac{2}{\sqrt[4]{3}}$, the second derivative is zero, so we must test values.

For a = ± 2 / $\sqrt[4]{3}$, critical value: $x = \pm\frac{1}{\sqrt[4]{3}} = \pm 0.76$, at which f(x) = 2.0944.

When x = ± 0.5: f(x) = 2.0917

When x = ± 1: f(x) = 2.0924 [1 mark]

These are smaller than the value at the stationary point so these are also **maximum** points. Therefore, in all cases, it is a maximum point. [1 mark]

Question 3

3a. f(x) is undefined when either the input to the log is zero, or there is a division by zero in the fraction (since log(infinity) diverges).

ln(0) occurs when: 1 + sin x = 0 → sin x = -1 → x = 3π/2, -π/2 [1 mark]
Division by zero occurs when: 1 - sin x = 0 → sin x = 1 → x = π/2, -3π/2 [1 mark]

The set is { -3π/2, -π/2, π/2, 3π/2 } = { ±π/2, ±3π/2 } [1 mark].

3b. We need to turn 1 and sin x into sec x and tan x. The easiest way to do this is to divide top and bottom by cos x:

$$f(x) = \frac{1}{2}\ln\frac{(1+\sin x)\div\cos x}{(1-\sin x)\div\cos x} = \frac{1}{2}\ln\frac{\sec x+\tan x}{\sec x-\tan x}$$

[2 marks; 1 for choosing to divide by cos x, 1 for getting the correct form]
Multiply top and bottom by the conjugate to create difference of squares:

$$= \frac{1}{2}\ln\frac{(\sec x+\tan x)\cdot(\sec x+\tan x)}{(\sec x-\tan x)\cdot(\sec x+\tan x)} = \frac{1}{2}\ln\frac{(\sec x+\tan x)^2}{\sec^2 x-\tan^2 x}$$ [2 marks]

Use the identity $\sec^2 x = 1 + \tan^2 x \rightarrow \sec^2 x - \tan^2 x = 1$:

$$= \frac{1}{2}\ln(\sec x+\tan x)^2$$ [1 mark]

Use the log identity, $\log x^a = a \log x$. The 1/2 cancels with the square, take the modulus to preserve the domain:

$$= \ln|\sec x+\tan x|.$$ [1 mark]

3c.

i) Starting with

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln|\sec x+\tan x|,$$

use chain rule: d/dx ln|x| = 1/x (no modulus), d/dx tan x = $\sec^2 x$

[1 mark for recalling and applying both these] and

d/dx sec x = d/dx 1 / cos x = sin x / $\cos^2 x$ = sec x tan x [1 mark], so

$$=\frac{\sec x\tan x+\sec^2 x}{\sec x+\tan x}=\frac{\sec x(\sec x+\tan x)}{\sec x+\tan x}=\sec x.$$ [1 mark]

ii) Differentiate g(x) using the product rule:

g'(x) = -cos x * f(x) + (1 - sin x) * f'(x)

= -cos x ln|sec x + tan x| + (1 - sin x) sec x [1 mark]

Transform x to x + π/2. This represents a shift of π/2 to the left.
We can visualise / quickly sketch the graphs to see that
cos(x + π/2) = -sin x, sec(π + π/2) = -csc x, tan(x + π/2) = -cot x and
sin(x + π/2) = cos x [4 marks]. Putting these in,

g'(x + π/2) = sin x ln|-csc x - cot x| - (1 - cos x) csc x

= (cos x - 1) csc x + sin x ln|csc x + cot x| [1 mark]

Now applying small-angle approximations,

sin x ≈ x, cos x ≈ 1 - $\frac{1}{2}x^2$, csc x = 1 / sin x ≈ cot x = 1 / tan x ≈ 1 / x,

g'(x + π/2) ≈ (1 - $\frac{1}{2}x^2$ - 1) (1 / x) + x ln|1 / x + 1 / x| [2 marks]

≈ -$\frac{1}{2}$ x + x ln|2 / x|. [1 mark]

iii) g(x) is undefined at x = π/2 (since from part a, f(x) was also undefined there). [1 mark]

Question 4

4a.

i) Write the expression in a form similar to the form where we can apply the binomial approximation formula:

$$\frac{\sqrt{1-x}}{\sqrt[3]{1+3x+x^2}} = (1-x)^{1/2} \cdot (1+x(x+3))^{-1/3}$$

[1 mark]

Expand the first factor using the formula:

$$(1-x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 + \cdots$$

[1 mark]

Similarly for the second factor, and including only terms up to x^2:

$$(1+x(x+3))^{-1/3} \approx 1 - \frac{1}{3}x(x+3) + \frac{2}{9}x^2(x+3)^2 + \cdots$$

[1 mark]

$$\approx 1 - x + \frac{5}{3}x^2 + \cdots$$

[1 mark]

Now multiply, again including only terms up to x^2:

$$(1-x)^{1/2} \cdot (1+x(x+3))^{-1/3} \approx \left(1 - \frac{1}{2}x - \frac{1}{8}x^2\right)\left(1 - x + \frac{5}{3}x^2\right)$$

$$\approx 1 - \frac{3}{2}x + \frac{49}{24}x^2 + \cdots$$

[3 marks]

ii) Since we used the binomial approximation twice, the overall valid interval will be the intersection of the intervals of each approximation.

For the first approximation, $(1 - x)^{1/2}$, interval is $|x| \leq 1$ [1 mark]

For the second approximation, $(1 + x(x + 3))^{-1/3}$, using the same process, the interval satisfies

$$x(x+3) \leq 1$$ [1 mark]

$$x^2 + 3x - 1 \leq 0$$ [1 mark]

$$-\frac{3+\sqrt{13}}{2} \leq x \leq \frac{\sqrt{13}-3}{2}$$ [1 mark]

The LHS is below -1, and the RHS is smaller than 1, so combining these two intervals, we get

$$-1 \leq x \leq \frac{\sqrt{13}-3}{2}.$$

However, notice that when x = -1, the denominator is zero so the function has an asymptote. Removing this point from the range, the interval is

$$-1 < x \leq \frac{\sqrt{13}-3}{2}$$ [1 mark]

4b.

i) Expand the numerator:

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x = \int_0^1 \frac{x^8-4x^7+6x^6-4x^5+x^4}{1+x^2}\,\mathrm{d}x$$

[2 marks]

Now applying long division,

$$\frac{x^8-4x^7+6x^6-4x^5+x^4}{1+x^2} = x^6-4x^5+5x^4-4x^2+4-\frac{4}{1+x^2}$$

[3 marks]

So the integral becomes (using the given result for d/dx $\tan^{-1} x$)

$$\int_0^1 x^6-4x^5+5x^4-4x^2+4-\frac{4}{1+x^2}\,\mathrm{d}x = \left[\frac{1}{7}x^7-\frac{2}{3}x^6+x^5-\frac{4}{3}x^3+4x-4\tan^{-1}x\right]_0^1$$

[2 mark]

Subbing in the numbers, using $\tan^{-1} 1 = \pi/4$,

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x = \frac{22}{7}-\pi$$

[1 mark]

ii) Since all the quantities in the integral: x^4, $(1 - x)$ and $1 + x^2$, are positive for $0 < x < 1$, the integral must be positive. [1 mark]

So $22/7 - \pi > 0$ and therefore $\pi < 22/7$. [1 mark]

Question 5

5a. Starting with $x^4 - 40x^2 + a = 0$, solve for x^2:

$$x^2 = \frac{40 \pm \sqrt{1600 - 4a}}{2} = 20 \pm \sqrt{400 - a}$$ [2 marks]

In order for there to be four real solutions, both of these values must be positive so then we can square-root them each to get x. We must also have this expression to be valid. Therefore

$20 - \sqrt{400 - a} \geq 0$ and $400 - a \geq 0$ [2 marks]

$400 - a \leq 400$ and $a \leq 400$

$a \geq 0$

Combining, the range of values of a is $0 \leq a \leq 400$. [2 marks]

5b. Writing out the four roots explicitly, they are (from smallest to largest):

$$x \in \left\{-\sqrt{20+\sqrt{400-a}}, -\sqrt{20-\sqrt{400-a}}, \sqrt{20-\sqrt{400-a}}, \sqrt{20+\sqrt{400-a}}\right\}$$

[2 marks]

It is clear that the first root and the last root are of equal magnitude and opposite sign, likewise for the second root and the third root. Therefore, these four roots must be equally distributed around x = 0.

Since the difference between the second and third (middle) roots must be the same as that between the third and fourth (positive) roots (arithmetic sequence), it follows that the last root is 3 times the third root. So:

$$3\sqrt{20-\sqrt{400-a}} = \sqrt{20+\sqrt{400-a}}$$

[5 marks]

Solve for a:

$$180 - 9\sqrt{400-a} = 20 + \sqrt{400-a}$$
$$160 = 10\sqrt{400-a}$$
$$a = 144$$

[2 marks]

5c.

i) If the roots were smallest to largest and geometric, then the ratios between consecutive terms would be positive, negative, positive. This cannot be any constant so cannot be geometric in this order. [1 mark]

ii) Consider the factored form using the roots.
Replace each root with A, Ar, Ar^2, Ar^3:

$$(x-A)(x-Ar)(x-Ar^2)(x-Ar^3) = x^4 - 40x^2 + a = 0$$ [2 marks]

The constant term in this expansion is equal to the product of the roots, i.e.
$(-A)(-Ar)(-Ar^2)(-Ar^3) = A^4r^6$. [2 marks]
Comparing to the constant term on the RHS, we get $a = r^6A^4$. [1 mark]

By inspection of the symmetry of the roots, the sum of the roots is 0 and so
$A + Ar + Ar^2 + Ar^3 = A(1 + r + r^2 + r^3) = 0$.
Using the partial sum of a geometric series, $A((r^4 - 1) / (r - 1)) = 0$
Since $A \neq 0$ and $r \neq 1$, we get $r^4 - 1 = 0 \rightarrow r = -1$. [2 marks]

iii) Putting these together, $a = A^4$. [1 mark]
It follows that the first and third terms (the two positive terms) are equal:

$$\sqrt{20 - \sqrt{400-a}} = \sqrt{20 + \sqrt{400-a}}$$ [2 marks]

Therefore, $\sqrt{400-a} = 0$ and so $a = 400$. [1 mark]

iv) There are actually only two roots, both of which are repeated. [1 mark]

Question 6

6a.

i) By 3D Pythagoras' theorem,

$$r = \sqrt{x^2 + y^2 + z^2}$$ [1 mark]

ii) From trigonometry in the right-triangle in the vertical plane:

$$\cos\theta = \sin(90^\circ - \theta) = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$ [2 marks]

iii) From trigonometry in the right-triangle in the horizontal plane:

$$\tan\phi = \frac{y}{x}$$ [1 mark]

6b.

i) The hypotenuse of the horizontal triangle is $r\sin\theta$ so by trig again,

$$x = r\sin\theta\cos\phi.$$ [2 marks]

ii) Combining with the result in 5a.iii), $y = r\sin\theta\sin\phi$. [1 mark]

iii) From trig in the vertical triangle, $z = r\cos\theta$. [1 mark]

6c. Starting from the given differential equation,

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}u}{\mathrm{d}r}\right)+\frac{1}{r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}=0$$

We now need to find all the derivatives of the given function. In each case, consider all other variables as constants:

$$u=\left(r+\frac{1}{r^2}\right)\sin\theta\cos\phi$$

$$\rightarrow\frac{\mathrm{d}u}{\mathrm{d}r}=\left(1-\frac{2}{r^3}\right)\sin\theta\cos\phi$$

$$\rightarrow\frac{\mathrm{d}u}{\mathrm{d}\theta}=\left(r+\frac{1}{r^2}\right)\cos\theta\cos\phi$$

$$\rightarrow\frac{\mathrm{d}u}{\mathrm{d}\phi}=-\left(r+\frac{1}{r^2}\right)\sin\theta\sin\phi\rightarrow\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}=-\left(r+\frac{1}{r^2}\right)\sin\theta\cos\phi$$

[3 marks]

Putting these into the differential equation, the LHS is

$$=\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left[\left(r^2-\frac{2}{r}\right)\sin\theta\cos\phi\right]+\frac{1}{r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left[\left(r+\frac{1}{r^2}\right)\sin\theta\cos\theta\cos\phi\right]$$
$$-\frac{1}{r^2\sin^2\theta}\left(r+\frac{1}{r^2}\right)\sin\theta\cos\phi$$

[1 mark]

Evaluating the derivatives, factoring out and expanding again carefully:

$$= \frac{2}{r^2}\left(r + \frac{1}{r^2}\right)\sin\theta\cos\phi + \frac{\cos\phi}{r^2\sin\theta}\left(r + \frac{1}{r^2}\right)\left(\cos^2\theta - \sin^2\theta\right) - \frac{\cos\phi}{r^2\sin\theta}\left(r + \frac{1}{r^2}\right)$$

$$= \left(r + \frac{1}{r^2}\right)\left[\frac{2\sin\theta\cos\phi}{r^2} + \frac{\cos\phi\cos^2\theta}{r^2\sin\theta} - \frac{\sin\theta\cos\phi}{r^2} - \frac{\cos\phi}{r^2\sin\theta}\right]$$

$$= \left(r + \frac{1}{r^2}\right)\left[\frac{\sin\theta\cos\phi}{r^2} + \frac{\cos\phi(\cos^2\theta - 1)}{r^2\sin\theta}\right]$$

[4 marks]

Apply identity $\sin^2\theta + \cos^2\theta = 1 \rightarrow \cos^2\theta - 1 = -\sin^2\theta$:

$$= \left(r + \frac{1}{r^2}\right)\left[\frac{\sin\theta\cos\phi}{r^2} - \frac{\sin\theta\cos\phi}{r^2}\right]$$

$$= 0 = \mathrm{RHS}.$$

[2 marks]

6d. Starting from

$$u = \left(r + \frac{1}{r^2}\right) \sin\theta \cos\phi$$

Putting in the known substitutions, deriving sin θ and cos φ again:

$$u = \left(\sqrt{x^2+y^2+z^2} + \frac{1}{x^2+y^2+z^2}\right) \cdot \sqrt{\frac{x^2+y^2}{x^2+y^2+z^2}} \cdot \frac{x}{\sqrt{x^2+y^2}}$$

$$u = x\left(1 + (x^2+y^2+z^2)^{-\frac{3}{2}}\right)$$

[2 marks]

The first derivatives, using chain/product rules, are

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 1 + (x^2+y^2+z^2)^{-\frac{3}{2}} - 3x^2\left(x^2+y^2+z^2\right)^{-\frac{5}{2}}$$

[4 marks]

$$\frac{\mathrm{d}u}{\mathrm{d}y} = -3xy\left(x^2+y^2+z^2\right)^{-\frac{5}{2}} \qquad \frac{\mathrm{d}u}{\mathrm{d}z} = -3xz\left(x^2+y^2+z^2\right)^{-\frac{5}{2}}$$

Similarly, the second derivatives, are

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} = \frac{15x^3 - 9x(x^2+y^2+z^2)}{(x^2+y^2+z^2)^{7/2}}$$

[3 marks]

$$\frac{\mathrm{d}^2u}{\mathrm{d}y^2} = \frac{-3x(x^2-4y^2+z^2)}{(x^2+y^2+z^2)^{7/2}} \qquad \frac{\mathrm{d}^2u}{\mathrm{d}z^2} = \frac{-3x(x^2+y^2-4z^2)}{(x^2+y^2+z^2)^{7/2}}$$

By inspection, the sum of the numerators is zero, so

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \frac{\mathrm{d}^2u}{\mathrm{d}y^2} + \frac{\mathrm{d}^2u}{\mathrm{d}z^2} = 0 .$$

[1 mark]

Question 7

7a. Multiplying through by dθ/dt:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\frac{\mathrm{d}\theta}{\mathrm{d}t}\sin\theta = 0$$

[1 mark]

In the first term, we see a function (dθ/dt) times its derivative ($d^2\theta/dt^2$). This seems similar to the chain rule. With a bit of thought we can see that if we differentiate $(d\theta/dt)^2$ we get 2 (dθ/dt) ($d^2\theta/dt^2$).

Following a similar process for the second term we can deduce that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2\right) + \frac{g}{L}\frac{\mathrm{d}}{\mathrm{d}t}\left(-\cos\theta\right) = 0$$

[2 marks]

Now integrate both sides with respect to *t*, we eliminate the d/d*t* and get a constant on the RHS (integral of 0 is 0*t* + *C*)

$$\frac{1}{2}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 - \frac{g}{L}\cos\theta = C$$

[1 mark]

7b. Initial conditions: $\theta = \theta_0$ and dθ/dt = 0 (since it was released from rest i.e. speed = angular speed = 0)

Putting this in to the LHS:
$\frac{1}{2} * 0^2 - g/L * \cos\theta_0 = C \rightarrow C = -g/L \cos\theta_0$ [1 mark]

Put this back in, rearrange and factorise:

$$\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{2g}{L}\left(\cos\theta - \cos\theta_0\right)$$

[1 mark]

7c. Applying separation of variables:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \sqrt{\frac{2g}{L}(\cos\theta - \cos\theta_0)}$$

$$\sqrt{\frac{L}{2g}}\frac{1}{\sqrt{\cos\theta - \cos\theta_0}}\,\mathrm{d}\theta = \mathrm{d}t$$

[2 marks]

Thinking about the motion of the pendulum, the time period will be the time taken to move through a complete oscillation: θ goes from θ_0, down through 0, up to $-\theta_0$, back down through 0 and up to θ_0 again. Since the times taken to do each of these is equal, we can say that $\frac{1}{4}$ of the time period is when the pendulum returns from 0 to θ_0 (to make the signs easiest).

Applying the integration, therefore

$$\sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{\cos\theta - \cos\theta_0}}\,\mathrm{d}\theta = \int_0^{T/4}\mathrm{d}t = \frac{T}{4}$$

[1 mark]

To use the substitution given in the question, we now need to convert cosines to half-angle sines. This can be done using the double angle formula:
$\cos 2x = 1 - 2\sin^2 x$: let $x = \frac{1}{2}\theta$ and $\frac{1}{2}\theta_0$ (separately):
$\rightarrow \cos\theta = 1 - 2\sin^2\frac{1}{2}\theta$ and $\cos\theta_0 = 1 - 2\sin^2\frac{1}{2}\theta_0$:

$$\frac{T}{4} = \sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{1 - 2\sin^2\frac{1}{2}\theta - 1 + 2\sin^2\frac{1}{2}\theta_0}}\,\mathrm{d}\theta$$

$$T = 4\sqrt{\frac{L}{2g}}\int_0^{\theta_0}\frac{1}{\sqrt{2\sin^2\frac{1}{2}\theta_0 - 2\sin^2\frac{1}{2}\theta}}\,\mathrm{d}\theta$$

[2 marks]

Use the given substitution, $\sin \frac{1}{2}\theta = \sin \frac{1}{2}\theta_0 \sin x$.
Differentiating implicitly with respect to x:
$\rightarrow \frac{1}{2}\cos \frac{1}{2}\theta * d\theta/dx = \sin \frac{1}{2}\theta_0 \cos x$
$\rightarrow d\theta = (2 \sin \frac{1}{2}\theta_0 \cos x) / (\cos \frac{1}{2}\theta)\, dx$ [1 mark]
Limits: $0 = \sin x \rightarrow x = 0$ and $\sin \frac{1}{2}\theta_0 = \sin \frac{1}{2}\theta_0 \sin x \rightarrow \sin x = 1 \rightarrow x = \pi/2$.

Applying these substitutions, we get

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2\sin\frac{1}{2}\theta_0 \cos x}{\cos\frac{1}{2}\theta\sqrt{2\sin^2\frac{1}{2}\theta_0 - 2\sin^2\frac{1}{2}\theta_0 \sin^2 x}}\, \mathrm{d}x$$

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2\cos x}{\cos\frac{1}{2}\theta\sqrt{2 - 2\sin^2 x}}\, \mathrm{d}x$$

[2 marks]

We now need to get rid of the θ term. Using the Pythagorean trig identity, $\cos^2 \frac{1}{2}\theta = 1 - \sin^2 \frac{1}{2}\theta = 1 - \sin^2 \frac{1}{2}\theta_0 \sin^2 x$. Put under square root:

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2\cos x}{\sqrt{(1 - \sin^2\frac{1}{2}\theta_0 \sin^2 x)(2 - 2\sin^2 x)}}\, \mathrm{d}x$$

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2\cos x}{\sqrt{2 - 2\sin^2 x - 2\sin^2\frac{1}{2}\theta_0 \sin^2 x + 2\sin^4 x \sin^2\frac{1}{2}\theta_0}}\, \mathrm{d}x$$

$$T = 4\sqrt{\frac{L}{2g}} \int_0^{\frac{1}{2}\pi} \frac{2\cos x}{\sqrt{2\cos^2 x - 2\sin^2 x \cos^2 x \sin^2\frac{1}{2}\theta_0}}\, \mathrm{d}x$$

$$T = 4\sqrt{\frac{L}{g}} \int_0^{\frac{1}{2}\pi} \frac{1}{\sqrt{1 - \sin^2 x \sin^2\frac{1}{2}\theta_0}}\, \mathrm{d}x$$

[4 marks]

7d.

i) Step size = 90/4 = 22.5° (π/8 radians).
The constant factor is $\sin^2 \frac{1}{2}\theta_0 = \sin^2 30 = \frac{1}{4}$ (to save time, use this)

Using trapezium rule on the integral part:

$$I = \frac{\pi}{16}\left(1 + 2\left(\frac{1}{\sqrt{1-\frac{1}{4}\sin^2\frac{\pi}{8}}} + \frac{1}{\sqrt{1-\frac{1}{4}\sin^2\frac{\pi}{4}}} + \frac{1}{\sqrt{1-\frac{1}{4}\sin^2\frac{3\pi}{8}}}\right) + \sqrt{\frac{4}{3}}\right)$$

I = 1.685750355 [3 marks]
→ T = 4 * √(0.5 / 9.81) * 1.685750355 = 1.522312276 [1 mark]

ii) Exact value = 4 * √(0.5 / 9.81) * 1.685750355 = 1.522312276 [1 mark]

(appears to be) exactly correct [1 mark]

Accuracy is extremely high / exact because the function is (almost) equally concave up as concave down on the interval so the error terms (almost) cancel out [1 mark]

Not the function is a straight line

(In fact the answer is not quite exact, the error is about 3 x 10^{-10}. However, since the calculator only displays 10 significant figures, the answer appears to match exactly.)